PRATISH DEWANGAN

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- Experienced **(5+ years)** in developing end to end Python Digital products and chatbot based or Agentic based application leveraging next gen technologies such as **Genai (LLM Model), Azure AI Cloud Integration, RAG, LangChain, LangGraph, Prompt Engineering**.
- Have hands on experience in **Automating the Deployment of ML Models** in Azure Machine Learning Studio by **Azure Devops CI/CD pipeline and also in GithubActions** with model registering and parameter monitoring in **ML Flow.** Have experience in building and training **NLP and Machine Learning Models like BERT, Random Forest, XgBoost** and Deploying it and also in Fine Tuning LLM Based models like **BERT base and LLAMA 2.0 34B parameters Model**.
- Have hands on experience **in Docker, Machine Learning Algorithms, writing scalable python code, tensor flow code and SQL queries, and have good proficiency in statistics,** and also have exposure to work in Angular and **Django Rest Frame work.**.
- Have Good exposure to Cloud services with **4X Microsoft** certified in **Azure AI Engineering Associate, Azure Data Scientist Associate.**

TECHNICAL SKILLS

Skills - MlOps in Azure Machine Learning Studio, Machine Learning, Deep Learning (NLP), RAG Model, GenAI, LLMs, Prompt Engineering, LLM Fine Tuning, LLMOps in (NVIDIA Nim, NVIDIA Neo)
Language – Python (Pandas, Matplotlib, Seaborn), TensorFlow and Pytorch, Lang Chain, Lang Graph
Techniques - Data Structure and Algorithms, Design Patterns, OPPS in Python
Azure - Azure AI Studio, Azure AI Search, Azure Machine Learning, Azure DevOps (CI/CD), Docker Registry, Azure Prompt Flow
FrameWork/Tools - Django REST Framework, Github Actions, Air Flow, DVC, Dashhub, ML Flow, Docker, AKS

EDUCATION

- Master of Technology in **Industrial Engineering & Operations Research from IIT Delhi** (2019-2021)
  Course Work – Supply Chain Management, **Operation Research, Stochastic Optimization**, Operation Plaining and Control, and more
- B.Tech in Mechanical Engineering from Government Engineering College (GEC), Raipur, CSVTU University (2015-2019)

WORK EXPERIENCE

Machine Learning Engineer | Tiger Analytics *[Nov 2024 – Present]*

Role – AI Observability | Pepsico

- Handling multiple AI Engineering projects such as CGPT (SQL Based Chatbot application), Gatorade-DID (RAG Based Chatbot application), and IT Control Automation (Agentic application). Doing **Instrumentation** (Integrating the application to Arize platform) **and Development** on these application that involves creating the custom evals, metrics, and dashboards in **Arize** for Observability and tracking the latency, Hallucinations, etc and fixing major issues with enhancing the existing feature.

Project – Manufacturing Defects / Pepsico

- Designed an end to end CV based manufacturing defects application. Its an **OCR based application** that detects the bellmark code, operator id and manuf time from the defects chips package and gives confidence score. It takes the images and download the image from Dataverse, process the OCR, and creates a RCA report and upload the results back to Dataverse. Added Arize Integration.
- Used Content Understanding from AI foundry as primary model and as a fall for Gpt 4.1 model for partial and failure Images. Added a part of classification where it classify the images as Bad and Good Images before processing. Build Automation script for Testing.

Impact – Reduction of 93% direct effort of manual entry of the bellmark codes to the dataverse.5% effort requires manual verification once.

Project – DataFlyWheel | Pepsico

- **Designed an end-to-end ML fine-tuning pipeline** for LLMs using **NVIDIA NeMo framework**, starting from **data curation pipeline** (including data extraction, PII masking, filtering, duplication) to fine-tuning using **NVIDIA NeMo microservices**.

Project – PowerBIChatBot | OSM THome

- Designed and implemented an **end-to-end ML pipeline** that powers a GenAI chatbot integrated with **PowerBI dashboards**, enabling natural language interaction with business KPIs and metrics.
- Used **GPT-4o mini** and a **RAG (Retrieval-Augmented Generation)** architecture to generate **DAX queries**, execute them over a semantic model, and summarize results using LLM-based **NLG (Natural Language Generation)**.
- Built an **agentic loop** for **auto-correction of DAX** with retry logic (up to 5 attempts), enabling a robust conversational interface for BI.
- Optimized total response time from 7s to 4s using **multi-threaded Python execution** and **asynchronous programming**.
- Developed **MLflow-integrated validation script** to evaluate accuracy of model-generated answers against user queries.
- Applied **prompt engineering**, **few-shot learning**, and domain-specific DAX examples to improve LLM performance to 100% response accuracy.
- Used **Microsoft Bot Framework** for chatbot UI, deployed on **Azure VM**, with **Docker** for containerization and **Cosmos DB** as persistent conversation storage.

Impact – Reduce the effort of manually giving the filters and analyzing the data for Low level worker who don't have access to PowerBI

AI/ML Engineer | HCL Tech *[Aug 2023-Oct 2024]*

Project – GenAI / HCL Internal

- Created an **end-to-end MLOps pipeline** to automate SDLC tasks from **user story generation to frontend/backend code development and cloud deployment**, using **LangChain**, **local LLMs**, and **Generative AI**.
- Integrated **LLAMA 2.0** and **Mistral MoE** for secure on-premise inference to meet enterprise data compliance requirements.
- Built multi-stage **ML pipelines** with **LangGraph** to manage code generation, validation, and testing in iterative loops for both backend (Python/Django) and frontend (Angular/TypeScript).
- Performed **instruction fine-tuning** on LLAMA 2.0 (34B) using **LoRA/QLoRA via PEFT** for custom application code generation.
- Developed containerized deployment flows using **Docker**, integrated with **Azure Web App** through **CI/CD pipelines** in **Azure DevOps**.

- Built a **RAG system** for real-time code context retrieval using **Azure AI Search** and indexed code versions for better LLM response quality.
- Used **MLflow** for experiment tracking, model performance monitoring, and storing metadata across multiple iterations.

*Impact – Reduce production effort of **backend code by 80%** and **frontend by 70%,** results in quick initialization of complete SDLC application with just a single prompt from user, & with predefined yaml settings result in **60%** reduction in effort in deploying the application in Azure Devops.*

Internal Tasks -

- Developed and deployed **NLP-based ML models** using **Azure ML Studio** with support for **hyperparameter tuning (Optuna)**, **pytest test suites**, logging, and monitoring using **MLflow**.
- Handled **CI/CD orchestration** using **GitHub Actions**, YAML scripting, and containerized model serving pipelines using **Docker**.

AI/ML Developer| HCL Tech *[Aug 2021-Jun 2023]*

Project – Genie / HCL Internal

- Built a full-scale **DataOps platform** for contextualization and transformation of upstream data, built on **Python** and **Django REST Framework**.
- Designed the full **ML lifecycle** from data ingestion to domain mapping using rule-based and ML-based fuzzy matching (Random Forest, XGBoost, LSTM, BERT).
- Integrated an NLP model pipeline using **transformer-based architectures** for matching user-uploaded datasets with domain-specific schema, achieving up to **94% accuracy**.
- Implemented a **custom rule engine** in Python for flexible transformation logic post domain mapping.
- Developed **Angular-based UI** and managed frontend-backend communication with TypeScript, ensuring seamless integration with backend ML logic.
- Designed file ingestion logic for **multiformat inputs (MF4, CSV, Excel)**, parsed and normalized for storage and processing.
- Containerized the backend components and models using **Docker** for easy deployment and scaling.

*Impact **–** Results in seamlessly handling of Upstream related Data. **Reduced 30%** effort in contextualizing & transforming the data for end D.S*

Internal Tasks –

Explored **Cognite Data Fusion (CDF)**, built **PowerBI dashboards** from CDF's industrial data, and applied **regression modeling** using **ensemble ML techniques** (Random Forest, LightGBM, GBoost).

PROJECTS

Discrete Event Simulation and Optimization Model of Kidney Transplantation | IIT DELHI THESIS *[Jul,2020-June 2021]*

- Collected data of Donor, Recipients, Cities, Hospitals from South ROTTO states and used the data to create a simulation model (of each Individual states as well as combination of whole south ROTTO region) which is used to find the following outputs: **probability of a patient receiving the organ within one to five years of registration**, average number of deaths per year due to lack of donated organs, average time to transplant for wait-listed patients.
- Performed **Hypothesis Testing (Chi-squared Test)** to evaluate probability distribution in data of Organ arrival and Patient arrival.
- Used **Stochastic Simulation Optimization** method to locate additional transplantation center in South ROTTO states to optimize parameters like average wait time, average transportation time and many more, a total of 20+ parameters involved.
- Coded entire simulation in Python from scratch have more than **10k lines of simulation codes**. Used Object Oriented Programming to create objects of Patient and Donor and to optimize reusable codes. Used central **global timeline approach** to facilitate combined simulation of 5 ROTTO states simultaneously.
- Created allocation policies with different combinations of CIT (Cold Ischemia time) radius, Nearest Transplantation Center **(Nearest Neighbor Method Applied)**, Max KAP score and analyzed the outcomes of each allocation policies impacted on the simulation outcomes.

GenAI OpenBox | Visual Studio Code Extension | Self-Initiated | 200+ Downloads *[Feb 2024]*

Build a Open Box type GenAi Co-Pilot Visual Code Extension in TypeScript where user can use their own API key to power this extension, it provides code generation and code summarization capabilities. Publish this extension into VS market place with name **GenAI OpenBox**

CERTIFICATIONS

Online Program -

- 2 Month **Software Engineering and Data Science Training | HCL Tech** *[Aug,2021-Oct,2021]*
- 2 Month **MLE** Foundation Training | **Tiger Analytics** *[Nov,2024-Jan,2025]*

ACHIEVEMENTS

- **Passed Azure Fundamental** Certification Exam (Azure AZ 900 certified) *[Feb-2023]*
- **Passed Azure AI Fundamental** Certification Exam (Azure AI-900 certified) *[Feb-2024]*
- **Passed Azure AI Engineer Associate** Certification Exam (Azure AI-102 certified) *[Mar -2024]*
- Passed **Azure Data Scientist Associate** Certification Exam (Azure DP-100 certified) *[Jul 2024]*
- Passed **Problem Solving Test in Python** | Hacker Rank *[2020]*
- Passed **International knowledge management** (IKM-Test in Python | *HCl* *[Mar,2022]*
- Earned **Gold star badges in Java, SQL and Python** (Hacker Rank) *[2021]]*
- **Graduate Aptitude Test in Engineering (GATE)**: Secured **98.46 Percentile** in GATE (Mechanical Engineering) *[2019]*